

A Proposal for a Multimodal Interactive Platform for Data Collection in Autism Play-Based Therapy Sessions

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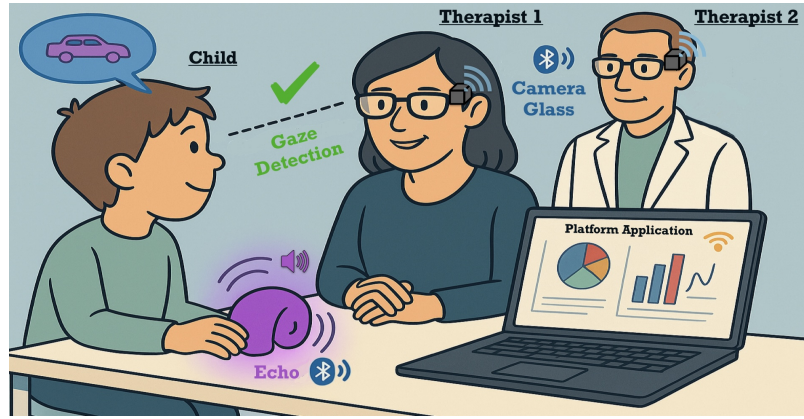


Figure 1: An overview of the proposed platform. The system collects synchronized behavioural data from two pairs of camera glasses and *Echo* – an interactive smart toy designed to stimulate symbolic play –, enabling users to visualize all data streams within an unified graphical user interface. This interface supports the analysis of behavioural patterns, offering a comprehensive perspective on the dynamic social interactions between child, therapists and toy.

Abstract

We present the prototype of a novel multimodal platform for behavioural data acquisition, designed to support therapists during play-based sessions with children with Autism Spectrum Conditions (ASC). Play-based therapy sessions can offer therapists valuable insights that support both diagnostic assessment and therapeutic decision-making. Since these type of insights come from several interactions between the therapists, the child and the environment; therapists could require automated multimodal systems that can capture and retain behavioural data beyond what can be observed through experience. The proposed system integrates four components: *Echo*, a smart toy capable to automatically recognize

different symbolic play activities when a child plays with it; 2 *Camera Glasses*, worn by two therapists, to record Video & Audio data from two different viewpoints; A *Computer Vision algorithm* to detect the gaze between the child and one of the therapist; and a *Platform Application* that allow to examine synchronized data logging to reconstruct the interaction between the child, the toy and the therapist within the session. The proposed environment is potentially a powerful time-saving data collection instrument for therapists and enables improved analysis to support research on social participation in ASC treatment.

CCS Concepts

• **Human-centered computing** → **Interaction design**; • **Applied computing** → *Health informatics*.

Keywords

Autism Spectrum Conditions; Assistive technology



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1 Introduction

Autism Spectrum Condition (ASC) is a neurodevelopmental disorder characterised by impaired or atypical behaviour in the social, emotional, and communicative domains [1]. These impairments typically emerge early in development and can have long-term effects on learning, social interaction, and personal autonomy. So, early-intervention (as early as 2 or 3 years of age) proved to be effective in reducing the core symptoms and improving, in the long term, the child's social competences [6, 12, 18, 22].

Play-based activities represent a key tool in therapeutic interventions for children with ASC [13]. Through different structured types of play, children can practice social behaviours, and express internal emotional states. Importantly, such activities provides observable behaviours that can provide therapists with meaningful information for assessment and treatment personalization [11, 21]. However, capturing and analysing all the relevant dimensions of a play based therapy session, such as the child's motor actions, attentional focus, relationship with the therapist, and emotional states, can be extremely challenging when relying only on therapist observation and experience.

To address this gap, recent years have seen the emergence of several technological devices and systems as a support to the therapists in play-based therapy sessions [3, 14]. In this context, our research group contributed by developing the idea of *Transitional Wearable Companions* (TWCs) [25]: interactive smart, soft toys which provide sensory feedback, specifically designed to support social engagement, while embedding sensing capabilities for behavioural data acquisition and monitoring. They include *PlusMe* [20], a panda conceived to stimulate imitation, and *X-8* [8], an octopus designed to stimulate turn-taking play. Lastly, we introduced *Echo*, the first AI-based TWC, which is designed to recognize symbolic play activities¹ [9]. While promising, these solutions mainly focus only on child-toy interaction, without integrating other relevant behavioural data.

A step in this direction was done in [19], which introduced a toy-based platform combining sensor data from *X-8* and gaze detection on video recorded from camera glasses worn by a therapist. This system was able to capture meaningful social interaction insights, such as joint attention² [4]. Building on this approach, which provided promising results, the current work aims to improve the previous system [19], by providing a time-saving multimodal synchronized data collection tool, extendable to all TWCs, based on a dual-camera setup and a visual synchronized platform (Figure 1). By allowing therapists to review play activities, gaze behaviour, and

audio/video data in a synchronized timeline, the system could supports the identification of social patterns - such as joint attention or avoidance - that may be crucial for therapy planning.

2 Related Work

In recent years, in addition to the aforementioned TWCs, several interactive technologies have been proposed to support children's therapy and learning through play, often integrating data collection mechanisms to better understand children's engagement and behaviour. Some examples are discussed in this section.

Beyond autism-specific applications, data-driven games have been explored as powerful tools for participatory sensing and user-centered design. For instance, *DataPet* by Dickinson et al. introduces a mobile game principally focused at children, using virtual pets to collect environmental data in a playful and participatory manner [7]. Similarly, Hidayat et al. used playful probes as method for gathering data from young users, supporting ideation and design refinement in game development [10].

Other efforts focus not only on finding the best game design for data acquisition, but also on behavioural monitoring and cognitive assessment. Pandria et al. present a series (5) of pilot studies using a series of digital platforms to track emotional states, behavioural changes, and attention traits (e.g., ADHD), showing how game-based systems can support diagnostic and therapeutic intervention [16]. Likewise, Nasuha et al. explored *Learning Analytics* the field AR-based games to monitor children's engagement and inform educational strategies through real-time performance data, showing benefits to children and educators [15].

Finally, Wilkie et al. have proposed an inspiring approach. It focuses on the development and validation of an observational data collection tool designed to assess physical literacy in children during motor games, by examining the dynamic interactions between the child and the environment. While no interactive gaming systems were employed, the use of video recording of motor games combined with software-based event tagging, enabled teachers to systematically collect behavioural data and monitor individual developmental trajectories over time. [23, 24].

These few examples highlight the growing interest in combining play, interaction, and data capture for both therapeutic and educational outcomes. However, most existing play-based data collection systems focus on augmenting play activities or collecting single-modal data streams outside of a clinical setting, without offering a comprehensive, synchronized structured environment tailored for therapeutic use with autistic children. Our work addresses this gap by proposing a more comprehensive data collection tool, that combines different synchronized data acquisition sources, allowing the user to record useful data both from a toy and from a structured environment, studying different types of behavioural interaction in the same play-based therapy session.

3 Platform Components

Figure 2 shows an overview of the components of the presented platform, that is composed by: 1) one of the previously presented TWC, *Echo*; 2) two *Camera Glasses* with embedded camera and microphone, the first supposed to be worn by a therapist near to the child during the play-based session and the second supposed to

¹A symbolic play activity is when a child use an object to imitate another object, and is a milestone in children communication skills development.

²Joint attention refers to a social behaviour where a child shares his/her attention about an object with another person. It is considered a foundational skill in early communication and social development.

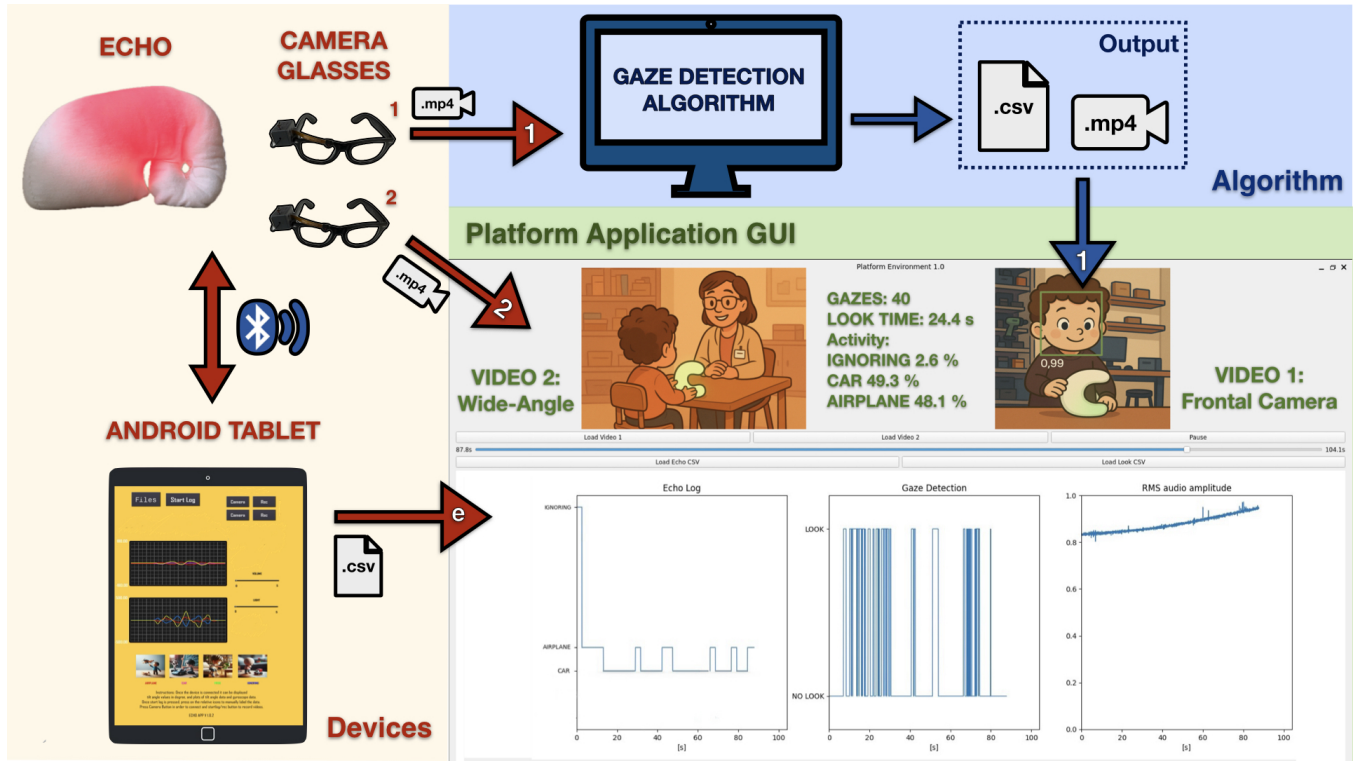


Figure 2: Platform Components. It is based on two data collection sources: the Camera Glasses (Audio and Video) & Echo. An Android App manage the Bluetooth connection and the synchronized data recording. Once the data are collected and processed, the user can visualize and navigate through them thanks to a Platform Application.

be worn by a second therapist that observe the scene (see Fig. 1); 3) an *Android App* that manages Bluetooth connections with the mentioned devices and file saving; 4) a *Computer Vision Algorithm* that process the recorded video in order to detect the gaze between the child and the first therapist; 5) a *Platform Application* where the therapist can load the data, visualize it in a synchronized way and receive insights about the play activity and gaze interaction.

First, we start with the description of *Echo* [2], the TWC used for the play-based therapy session. *Echo* is a soft, smart toy prototype developed to stimulate symbolic play activities in autistic children. It can recognize different symbolic play activities (e.g. playing with it by emulating an airplane or a car, or simply ignoring it) as achieved an average accuracy of 96% on a pilot test, thanks to an embedded, real-time Machine Learning algorithm and an IMU sensor. The toy is designed to be easily graspable, encouraging natural interaction. Once an activity is recognized, *Echo* emits light and sound feedback to reinforce the child's symbolic play, and logs each recognized action with a timestamp for later analysis. Alongside with this toy, an *Android App* was developed to allow the user to: Bluetooth pair with *Echo* and the Camera Glasses; visualize the sensor data; save a log file of the entire session, containing the recognized symbolic game activity. This entire system allowed us to synchronize the start and the stop of the toy data recording with video and sound data coming from the two different glasses.

The *Camera Glasses* prototype is shown in Figure 3. It is com-

posed by a little 3D printed case that can be mounted in any commercial glasses thanks to its little shape. The hardware is based on a *XIAO ESP32S3 Sense Module* board (fabricated by Seed Studio) and a commercial *OV2640-75mm* camera, all powered by a 3.7 V Li-Po Battery. This little board allowed us to manage Bluetooth connection with the android app and mostly to save the recorded video and audio session data on a microSD memory card, with a resolution of 800x600 (SVGA) at 20 fps stable. Two of these prototype were developed and they should be worn by the therapists during the play-based therapy session. One of the therapist should wear it near the child during the play activity to record gaze & joint attention data, while for the other therapist we have chosen a wide-angle camera to record the entire environment in a backward position. This second scene in particular, could be also used in the future to train an already developed Machine Learning-based event detection tool based on images collected during TWC-based therapy sessions [17]. The versatility of the whole system, furthermore, offers in the future, the possibility to connect more cameras or more, different devices like others TWCs, in order to easily collect – in a synchronized way – as much as possible behavioural data.

To obtain data on the child's joint attention and gaze, we processed the video of the of the first camera with an already developed *Computer Vision Algorithm*, which detects whether the child is looking at the therapist (i.e., Gaze Detection) [5]. The system is based on a recurrent architecture that combines Convolutional Neural

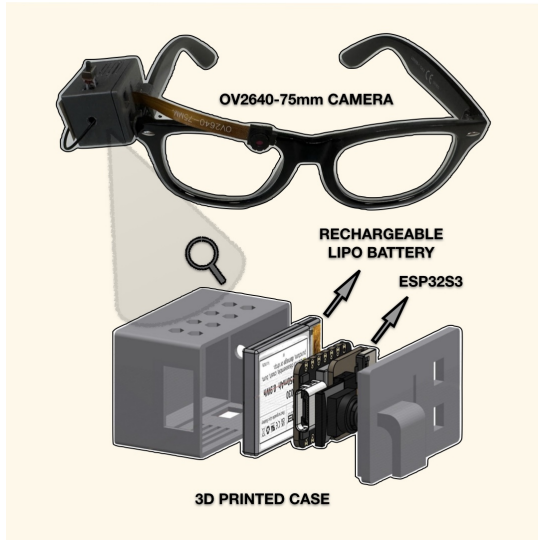


Figure 3: Inside the Camera Glass device. The simple shape of the case allows easy installation on a pair of glasses or to use it as an independent camera. The ESP32S3 module manages Bluetooth connection with the Android Application and Video & Audio recording.

Networks (CNNs) for frame-by-frame visual feature extraction and Long Short-Term Memory (LSTM) networks to model the temporal dynamics of eye contact. The deep learning model achieved expert-level accuracy, with ROC AUC scores exceeding 0.80, supporting its use in clinical settings for monitoring social attention in children. For our purposes, we used this algorithm to process the first video, to provide a post-processed version of the video, showing when the child’s gaze is detected with a coloured square (green or red) surrounding the face³. This output is also produced as a log file.

Once all videos, audio and log data are collected, they can be uploaded in our Python-based *Platform Application* (see Figure 2). A video explanation of the *Platform Application* usage after a test session, which supports the next description, is provided⁴. Our application’s GUI provides an integrated environment for the synchronized visualization and analysis of the data recorded during the play-based therapy session. The top section of the interface displays the two recorded video streams: the first (right) shows the therapist’s point of view near the child, including gaze detection; the second (left) provides a wide-angle view of the scene captured by the second camera, offering contextual information on the overall interaction. A timeline slider allows the user to navigate through the session synchronously. The bottom section of the interface is divided into three synchronized plots: (1) *Echo*’s Log shows the symbolic play activity performed by the child during the session (e.g., AIRPLANE, CAR or IGNORING); (2) the Gaze Detection plot illustrates whether the child was looking or not at the therapist during the session; (3) the Audio Waveform graph represents the audio intensity extracted from the environmental video, which can

help identify vocalizations or significant acoustic events during the session. Additionally, on the central part, summary statistics are automatically displayed, including the total number of detected gazes, cumulative look time, and the percentage distribution of symbolic play activities detected. While simple, these statistics provide important indicators of child social behaviour (for example how much the child interact with the toy and the therapist), and in the future can be expanded by adding other metrics linked to the collected data, such as joint attention events. This comprehensive interface allows therapists and researchers to explore play-based therapy sessions in a structured and data-driven manner. Typically, one therapist engages with the child using *Echo* while wearing the first pair of *Camera Glasses*, and a second therapist records from a wider perspective. After the session, all synchronized data streams can be reviewed to identify key interaction moments—such as symbolic play patterns and eye contact episodes—that would otherwise require time-consuming manual annotation or remain unnoticed, ultimately supporting clinical evaluation and decision-making.

4 Conclusion & Future Works

In this paper, we presented a novel multimodal interactive platform specifically designed to support play-based therapeutic sessions for children with Autism Spectrum Conditions (ASC). By integrating *Echo*, a smart toy capable of recognizing symbolic play, two Camera Glasses, a Gaze Detection algorithm, and a Platform Application for data visualization, our system offers a comprehensive solution to enrich therapeutic practices with structured behavioural data. The platform – once tested in a pilot experiment – will enable therapists to access synchronized views of child’s behaviour, gaze detection and interaction with the toy, potentially supporting an objective evaluation of social engagement and his/her developmental progress.

Technical future developments will focus on: i) Camera Glasses hardware limitations, such as extending battery life (now is about 25 minutes) and increasing video quality; ii) Platform Software limitations, since the current system still relies on a manual workflow for saving, processing, and uploading data; our next step is to deploy a safe, cloud-based infrastructure that will automate these processes and ensure data is stored, organized, and accessed in a structured, scalable, and compliant environment.

From a clinical point of view, the validation of the proposal will require, as a next step, the design of a structured experimental protocol, developed in collaboration with neurodevelopmental therapists, to assess its usability, effectiveness, and contribution to therapeutic outcomes in clinical environments. In this sense, the versatility of the proposed platform, in addition to allowing integration with new devices and systems (e.g., other TWCs or cameras), allows it to be shaped according to the needs of therapists.

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³See an example at: https://im-twin.eu/video/#eye_contact_detector_pilot_test

⁴See Online Supplementary Material.

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